

Time-Series Agent system

A module-level construction view

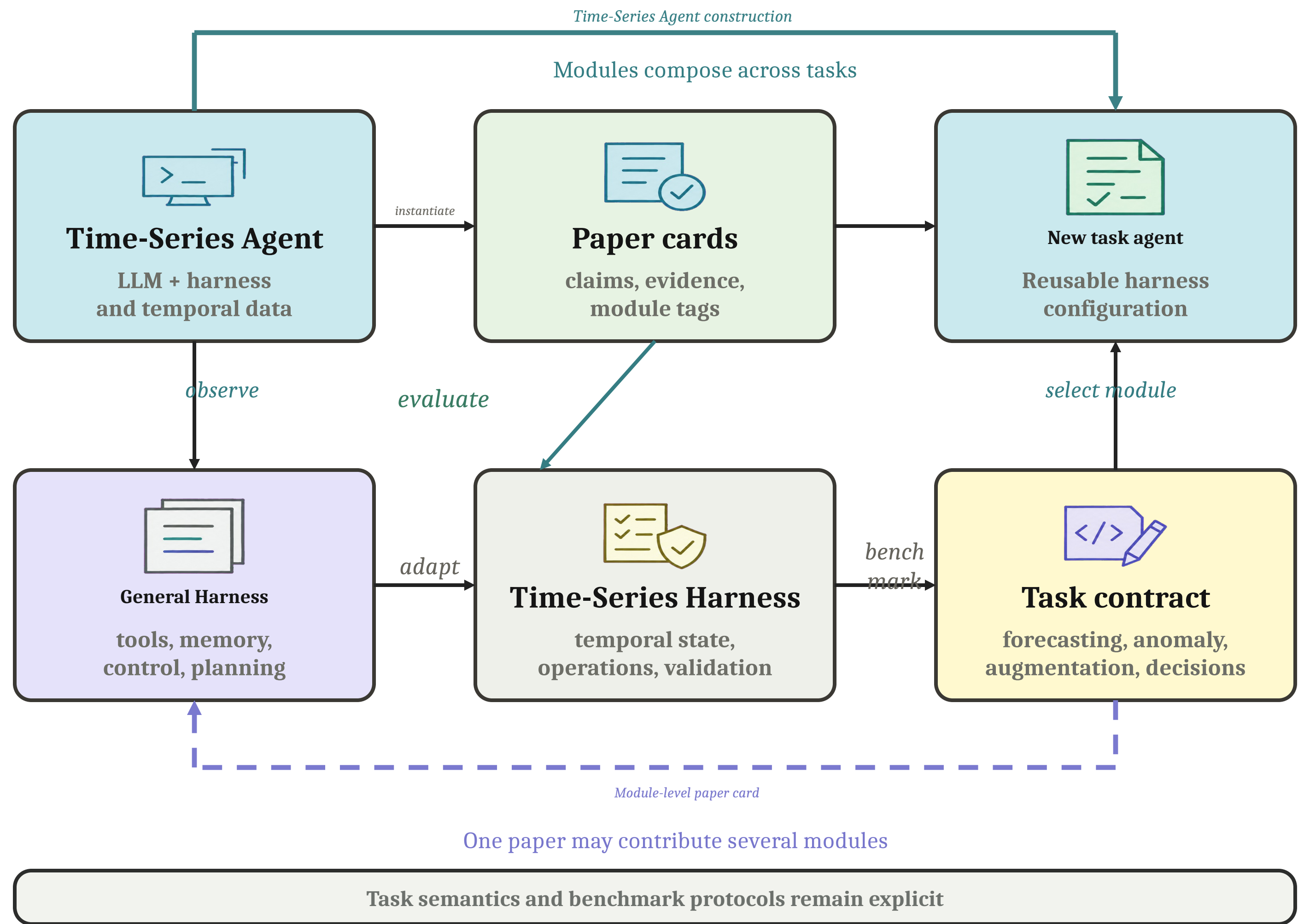


Figure 1. A module-level construction view of a Time-Series Agent. The LLM supplies language and reasoning; the General Harness provides reusable runtime mechanisms; the Time-Series Harness Module supplies temporal state, operations, grounding, and task contracts. A paper card records the claim, evidence locator, module layer, task family, and BibTeX entry.

Time-Series Agent

Reader's guide

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§ 1 Introduction	System definition Construction questions
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§ 8 Open Problems and Case Studies	Paper-card database Open tasks Module-level counts Evidence gaps
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The Time-Series Agent landscape

LLM, Harness modules, task contracts, and evaluation infrastructure

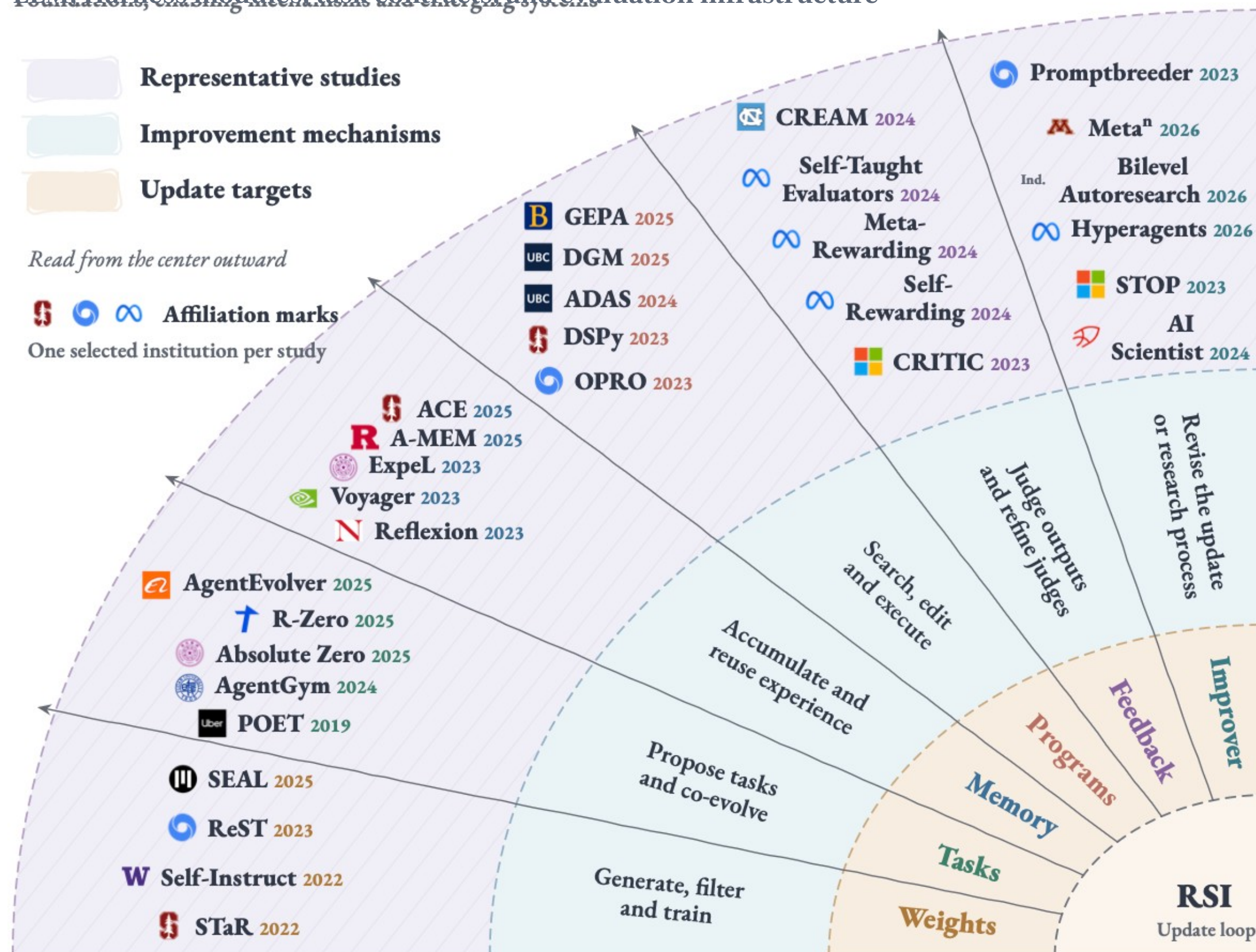


Figure 3. Landscape of mechanisms relevant to Time-Series Agents. The map connects the LLM side, reusable Harness modules, temporal

The inner band denotes the four contribution layers; the middle band denotes reusable mechanisms; the outer band denotes representative works.

The sectors cover language capability, runtime control, temporal grounding, task contracts, evaluation, and paper-card evidence. A work may contribute to several sectors; its placement highlights one module-level connection. The map includes LLM-side methods, Harness modules, time-series systems, and benchmark infrastructure. Logos denote selected affiliations or project homes, rather than base models; other collaborating institutions may be omitted. The accompanying paper-card register records each selection.

Research routes in Time-Series Agent systems

Representative module contributions by first public year | LLM and Harness layers

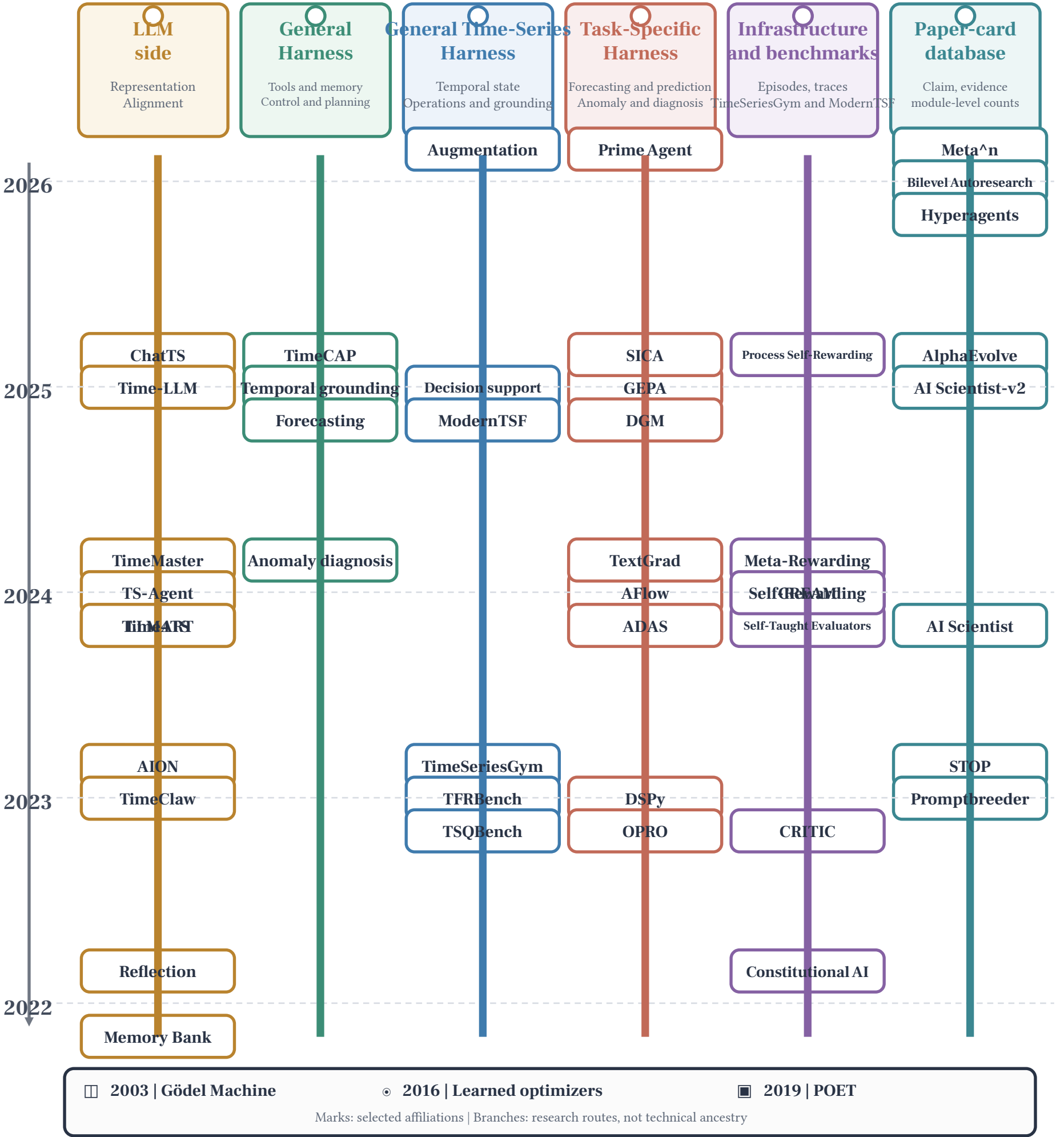


Figure 4. Research routes toward recursive self-improvement. Forty-five representative method papers are grouped into six research routes, with three historical anchors shown separately.

Year bands use the first public dates in the checked source records; spacing within each band is adjusted for legibility. Branch lines denote research families.

Technical ancestry, cross-route transfer, and sustained recursive gains require separate evidence. The selection reflects the literature search through September 12, 2026.

Logos identify one selected author or work-done affiliation per study; “Ind.” denotes independent researchers. Affiliation and logo sources are documented separately.

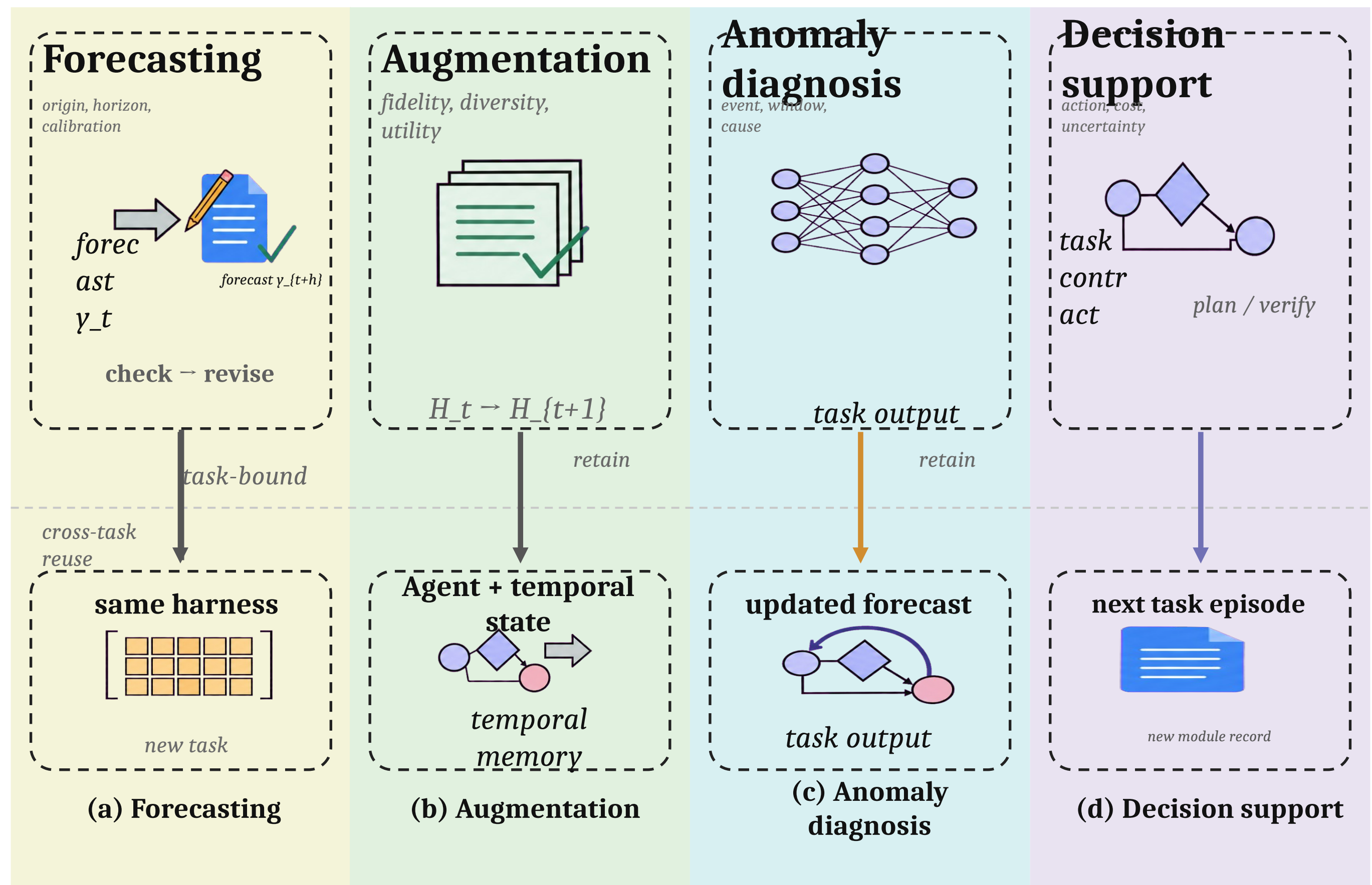


Figure 5. Four task contracts instantiated by Time-Series Harness Modules. The same general runtime can support distinct temporal outputs when the interface, memory, tools, control, planning, optimization, and coordination obligations are specialized.

Forecasting binds origin, horizon, and calibration. Augmentation binds fidelity, diversity, and utility. Anomaly diagnosis binds event windows and causes. Decision support binds actions, costs, and uncertainty.

A paper may instantiate more than one contract. The paper-card ledger records each module independently so that a task-specific contribution does not occupy a coarser layer.

The dashed horizontal rule separates reusable harness mechanisms from task-bound contracts.

Colored arrows show the output contract carried by each task family.

The visual vocabulary is adapted from the supplied RSI template; the content follows our four task families.